VIETNAM NATIONAL UNIVERSITY, HO CHI MINH CITY

HO CHI MINH CITY UNIVERSITY OF TECHNOLOGY

Course Project Report: EMBEDDED SYSTEM

Topic: MP3 BLUETOOTH PLAYBACK SYSTEM

Class: CC01

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ABSTRACT

This report presents the design and implementation of an MP3 Bluetooth Playback System using the STM32F411CEU6 microcontroller as the main controller. The system is designed as a portable audio player that supports two playback modes: MP3 playback from a microSD card through a DFPlayer Mini module and wireless Bluetooth playback through a Bluetooth audio receiver module. It also includes an OLED display, button-based control, audio source switching, speaker output, 3.5 mm audio jack output, and portable power operation using either a 12 V adapter or a 3S Li-ion battery pack.

The project follows a complete embedded system design process, including requirement analysis, component selection, schematic design, firmware organization, PCB layout, Proteus simulation, and breadboard prototyping. The hardware is divided into functional blocks, while the firmware is organized into separate modules for input handling, playback control, OLED rendering, battery monitoring, and application-level event processing.

The system was verified through simulation and breadboard testing. The prototype demonstrated the main functions, including playback control, OLED status display, button interaction, audio source selection, and speaker output. Some limitations remain, especially audio noise and the need for complete PCB-level testing after assembly.

CHAPTER 1. INTRODUCTION

1.1 Project Overview

Audio playback systems are practical examples of embedded systems because they require the integration of hardware, firmware, power management, user interface, and signal routing. A simple portable audio player may appear to be a basic electronic device, but its design involves many important embedded system concepts, such as microcontroller control, serial communication, button input processing, display control, audio source selection, voltage regulation, and PCB layout.

This project focuses on the design and implementation of an MP3 Bluetooth Playback System using the STM32F411CEU6 microcontroller. The system supports two audio playback modes. In MP3 mode, audio files are read from a microSD card and played through a DFPlayer Mini module. In Bluetooth mode, audio is received from a paired Bluetooth device through a Bluetooth audio receiver module. The selected audio source is routed to an audio amplifier circuit and then output through a speaker or a 3.5 mm audio jack.

The system also includes an OLED display for showing playback and system status, physical push buttons for user control, and a power supply circuit that supports both adapter and battery operation. The complete project includes schematic design, PCB layout, firmware development, Proteus simulation, breadboard prototyping, and PCB fabrication.

1.2 Motivation

The motivation of this project is to apply embedded system design knowledge to a complete and practical electronic product. Instead of developing only a simple microcontroller program, this project requires both hardware and firmware design. The system must combine several hardware modules and circuits into one working device, while the firmware must coordinate user input, playback control, display update, and system monitoring.

This project is also suitable for an embedded system design course because it covers many common design tasks. On the hardware side, the project includes power conversion, voltage regulation, audio routing, amplifier design, button input, OLED interface, and PCB layout. On the software side, the project includes peripheral configuration, modular C code organization, event-based button handling, display rendering, playback control, and battery monitoring.

Through this project, the group can gain practical experience in the full embedded development process, from requirement definition and component selection to schematic design, simulation, prototype testing, and PCB implementation. These skills are important because real embedded systems require not only correct code but also stable hardware, proper power design, good signal routing, and careful system-level testing.

1.3 Objectives

The main objective of this project is to design and implement a portable MP3 Bluetooth playback system based on the STM32F411CEU6 microcontroller.

1. To design a complete schematic for the MP3 Bluetooth playback system, including the power supply, STM32 main controller, audio source modules, audio source switching circuit, amplifier circuit, OLED display, push buttons, battery monitoring, and programming/debugging interface.
2. To select suitable components and modules for the system, including the STM32F411CEU6 microcontroller, DFPlayer Mini module, Bluetooth audio receiver module, OLED display, CD4053BE analog multiplexer, LM386 audio amplifier, voltage regulators, switches, connectors, and passive components.
3. To develop a modular firmware structure that separates the main application logic from input handling, playback control, OLED display rendering, and battery monitoring.
4. To implement user control functions such as mode selection, play/pause, next/previous track, and volume adjustment through physical buttons.

5. To simulate the system behavior in Proteus in order to verify the firmware flow and peripheral interaction before hardware implementation.
6. To build a breadboard prototype to validate the main system functions before finalizing the PCB.
7. To design and fabricate a PCB that integrates the complete system into a compact hardware platform.
8. To evaluate the current system performance, identify existing limitations, and propose future improvements, especially in audio noise reduction and PCB-level validation.

1.4 Scope and Limitations

The scope of this project includes the system-level design, simulation, prototyping, and implementation of a microcontroller-based MP3 Bluetooth playback system. The project focuses on integrating existing audio modules and supporting circuits into a complete embedded device. The STM32F411CEU6 microcontroller is used as the central controller, while MP3 decoding and Bluetooth audio reception are handled by external modules.

The project covers the following main functions: MP3 playback, Bluetooth playback, audio source switching, button-based user control, OLED status display, speaker output, 3.5 mm audio output, battery monitoring, and dual power operation. It also includes schematic design, PCB layout, modular firmware design, Proteus simulation, breadboard testing, and PCB fabrication.

However, the project also has several limitations. The audio quality is mainly evaluated through practical listening tests rather than professional measurement equipment. Some audio noise is still present in the prototype and needs further improvement. The PCB has been fabricated, but full PCB-level validation still needs to be completed after soldering and inspection. In addition, advanced features such as enclosure design, battery charging, advanced audio filtering, and detailed audio performance measurement are outside the main scope of the current implementation.

CHAPTER 2. REQUIREMENTS AND ARCHITECTURE

2.1 System Scope

The MP3 Bluetooth Playback System is designed to provide flexible audio playback from two main sources: MP3 files stored on a microSD card and Bluetooth audio from a paired device. The system allows the user to switch between MP3 mode and Bluetooth mode, control playback using physical buttons, and view the current system status on an OLED display. The output audio can be played through an onboard speaker or a 3.5 mm audio jack.

The system is intended to operate as a compact and portable embedded audio device. It supports power input from either a 12 V adapter or a 3S Li-ion battery pack. The power supply circuit generates the required voltage rails for the STM32 microcontroller, audio modules, OLED display, and amplifier circuit. At the system level, the design combines digital control, analog audio routing, power management, and user interface functions into one integrated platform.

Although the initial requirement document defined the early functional and non-functional requirements for the project, some details were later adjusted during schematic design and prototype development. Therefore, the requirements in this report are rewritten based on the final system architecture and actual implementation direction.

2.2 System Requirements

2.2.1 Functional Requirements

The main functional requirements of the MP3 Bluetooth Playback System are summarized in Table 2.1.

Requirement ID	Requirement Name	Description
FR-1	System Startup	The system shall initialize all required hardware modules after power-up, including the STM32, audio modules, OLED display, buttons, and battery monitoring circuit.
FR-2	Mode Selection	The system shall support at least two playback modes: MP3 mode and Bluetooth mode.
FR-3	MP3 Playback	In MP3 mode, the system shall play MP3 files from a microSD card through the DFPlayer Mini module.
FR-4	Bluetooth Playback	In Bluetooth mode, the system shall receive and play audio from a paired Bluetooth device.
FR-5	User Control	The system shall allow the user to control playback through physical buttons, including play/pause, next track, previous track, volume adjustment, and mode selection.
FR-6	OLED Display	The system shall display important information such as current mode, playback status, track information, volume level, and battery status on the OLED screen.
FR-7	Audio Source Switching	The system shall switch between MP3 audio and Bluetooth audio using an audio source selection circuit controlled by the MCU.

FR-8	Audio Output	The system shall provide audio output through a speaker and a 3.5 mm audio jack.
FR-9	Battery Monitoring	The system shall monitor the battery voltage using the ADC input of the STM32 microcontroller.
FR-10	Programming and Debugging	The system shall provide a programming and debugging interface through SWD for use with ST-Link.

Table 2.1 Functional requirements

2.2.2 Non-functional Requirements

Table 2.2 presents the main non-functional requirements related to portability, reliability, power stability, audio quality, firmware maintainability, and PCB implementation.

Requirement ID	**Requirement Name**	**Description**
NFR-1	Portability	The system should be compact enough to be used as a portable audio playback device.
NFR-2	Stable Power Operation	The system should generate stable voltage rails for the MCU, audio modules, display, and amplifier circuit.
NFR-3	Audio Quality	The audio output should be clear enough for normal listening, and background noise should be minimized as much as possible.
NFR-4	Responsiveness	The user interface should respond quickly to button input during normal operation.

NFR-5	Reliability	The system should operate without unexpected reset or malfunction during normal adapter-powered or battery-powered operation.
NFR-6	Maintainable Firmware	The firmware should be organized into separate modules to improve readability, debugging, and future extension.
NFR-7	PCB Manufacturability	The PCB should be suitable for fabrication, soldering, inspection, and testing.
NFR-8	Noise Reduction	The PCB layout should consider grounding, decoupling, and audio trace routing to reduce power and audio noise.
NFR-9	Expandability	The design should allow future improvements such as better battery management, improved audio filtering, enhanced user interface, and enclosure design.

Table 2.2 Non-functional requirements

2.3 System Architecture

The MP3 Bluetooth Playback System is organized as a modular embedded system, with the STM32F411CEU6 microcontroller acting as the central controller. The system receives audio from two sources: the DFPlayer Mini module for MP3 playback from a microSD card and the Bluetooth audio receiver module for wireless playback. The selected audio signal is routed through an analog source selection circuit before being sent to the amplifier and output stage.

The system also includes several supporting blocks. The OLED display provides visual feedback to the user, while physical push buttons are used for playback control and mode selection. The power supply circuit supports operation from either a 12 V adapter or a 3S Li-ion battery pack. The input voltage is converted into the required voltage rails for the MCU, audio modules, display, and amplifier circuit. In addition, the battery voltage is monitored through an ADC input of the STM32 microcontroller.

This architecture separates the system into clear functional blocks. Each block has a specific role, which makes the schematic easier to understand, the PCB easier to organize, and the firmware easier to develop and debug.

2.3.1 Overall Block Diagram

Figure 2.1 shows the overall block diagram of the MP3 Bluetooth Playback System. The STM32F411CEU6 is placed at the center of the system and communicates with different peripherals through UART, I2C, GPIO, ADC, and control signals.

The DFPlayer Mini module is connected to the STM32 through UART. It is responsible for reading and playing MP3 files from a microSD card. The Bluetooth receiver module provides wireless audio input from an external device such as a smartphone. The audio outputs from the DFPlayer Mini and Bluetooth module are connected to an analog audio switching circuit. The STM32 controls this switching circuit to select the active playback source.

The OLED display is connected to the STM32 through the I2C interface and is used to show system information such as playback mode, track number, volume level, and battery status. Push buttons are connected to GPIO pins and are used for user input, including play/pause, next track, previous track, volume adjustment, and mode selection. The selected audio signal is amplified by the LM386 amplifier circuit and then output through a speaker or a 3.5 mm audio jack.

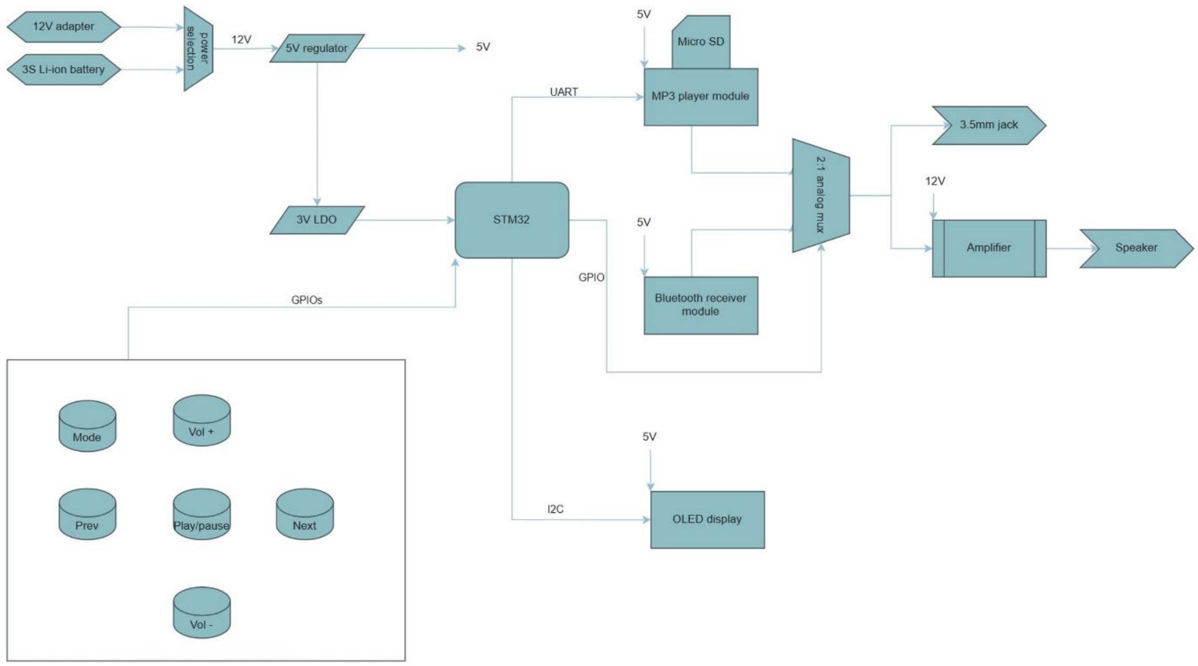

Figure 2.1. Overall system block diagram

2.3.2 Power Tree

The power tree of the system is shown in Figure 2.2. The system is designed to operate from two possible power sources: a 12 V adapter or a 3S Li-ion battery pack. These power sources provide the main input voltage for the system. The input voltage is then regulated to generate the voltage rails required by different blocks.

The main 5 V rail is generated using a DC-DC converter. This rail supplies the audio modules, OLED display, analog multiplexer, and other 5 V-compatible circuits. A 3.3 V rail is generated using a voltage regulator to supply the STM32F411CEU6 and other low-voltage logic circuits. Proper voltage regulation is important because unstable power can cause MCU reset, display malfunction, audio distortion, or unstable module operation.

The power design also considers current demand from the audio amplifier and audio modules. Since the amplifier and speaker can draw relatively high current during playback, the power supply circuit must provide enough current margin. Decoupling capacitors are placed near important components to reduce voltage fluctuation and improve system stability.

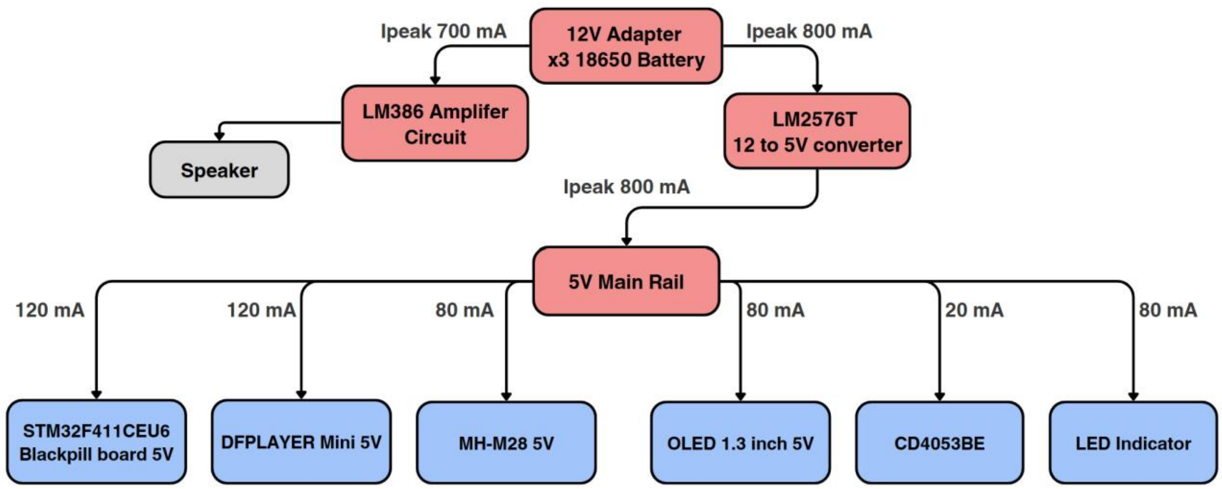

Figure 2.2. System power tree

2.4 Design Preparation

Before the schematic and PCB design were finalized, the main components, controller, and development tools were selected based on the system requirements. The design preparation stage helps ensure that each selected component is suitable for the required function and can be integrated into the complete system.

The main considerations in this stage include the number of required interfaces, operating voltage, module availability, ease of firmware development, PCB integration, and testing method. Since the project focuses on embedded system integration, several ready-made modules were used for audio playback and Bluetooth reception. This allows the project to focus more on system design, firmware control, audio routing, power design, PCB layout, and hardware testing.

2.4.1 MCU Selection

The STM32F411CEU6 microcontroller was selected as the main controller of the system. It is responsible for coordinating the operation of all major blocks, including MP3 playback control, OLED display update, button input handling, audio source selection, and battery monitoring.

The STM32F411CEU6 is suitable for this project because it provides the required peripheral interfaces. UART is used to communicate with the DFPlayer Mini module. I2C is used to control the OLED display. ADC is used to measure the battery voltage. GPIO and external interrupt lines are used to detect button inputs. The MCU also supports SWD programming and debugging through ST-Link, which makes firmware development and testing more convenient.

Another reason for selecting the STM32F411CEU6 is that the project does not require the MCU to directly decode MP3 audio or process Bluetooth audio data. These tasks are handled by external modules. Therefore, the MCU mainly performs control and coordination tasks, which are well within the capability of the STM32F411CEU6.

Selection Factor	**Reason**
UART support	Required for controlling the DFPlayer Mini module.
I2C support	Required for communicating with the OLED display.
ADC support	Required for battery voltage monitoring.
GPIO and EXTI	Required for button input and event detection.
SWD interface	Allows programming and debugging using ST-Link.
STM32CubeIDE support	Simplifies firmware development and peripheral configuration.
Processing capability	Sufficient for control logic, user interface, and system management.
Compact size	Suitable for integration into a portable PCB design.

Table 2.3 Microcontroller Selection Criteria

2.4.2 Component Selection

The components were selected based on the required system functions, availability, ease of integration, and compatibility with the STM32 controller.

The DFPlayer Mini was selected for MP3 playback. It supports MP3 files stored on a microSD card and can be controlled through UART commands. Using this module reduces firmware complexity because the STM32 does not need to decode MP3 files directly.

The Bluetooth audio receiver module was selected to provide wireless audio playback. It receives audio from a paired device and provides an analog audio output. This output can be routed to the amplifier through the audio source switching circuit.

The OLED display was selected as the user interface display. It is compact, low power, and can be controlled through I2C. It is used to show the current playback mode, playback status, volume level, and battery status.

The CD4053BE analog multiplexer was selected for audio source switching. It allows the STM32 to choose whether the amplifier receives audio from the DFPlayer Mini or from the Bluetooth module.

The LM386 audio amplifier was selected for the speaker output stage. It is a simple and commonly used low-power audio amplifier suitable for small speaker applications.

The DC-DC converter and voltage regulator circuits were selected to generate stable supply rails from the adapter or battery input. The 5 V rail supplies most audio modules and peripheral circuits, while the 3.3 V rail supplies the STM32 and low-voltage logic.

Component	Purpose in System
STM32F411CEU6	Main controller for system logic and peripheral control.
DFPlayer Mini	MP3 playback from microSD card.

Bluetooth audio receiver module	Wireless Bluetooth audio input.
OLED display	Displays mode, status, volume, and battery information.
Push buttons	User input for playback and mode control.
CD4053BE	Analog audio source selection.
LM386	Audio amplification for speaker output.
DC-DC converter	Generates the main 5 V rail.
3.3 V regulator	Supplies STM32 and low-voltage logic circuits.
ST-Link/SWD header	Programming and debugging interface.

Table 2.4 Selected Components and Their Functions in the System

2.4.3 Development Tools

Several tools were used during the development process. Each tool supports a different stage of the embedded system design flow, from hardware design to firmware development and testing.

Altium Designer was used for schematic design and PCB layout. The schematic was divided into functional blocks, including power supply, STM32 controller, audio source, audio routing, amplifier, display, buttons, and programming interface. The PCB layout was also designed in Altium, with attention to component placement, power routing, audio routing, grounding, and decoupling.

STM32CubeIDE was used for firmware development. It was used to configure MCU peripherals, write application code, compile the firmware, and support debugging.

ST-Link was used to program and debug the STM32F411CEU6 through the SWD interface. This allows the firmware to be tested directly on the hardware.

Proteus was used for simulation. The simulation helped verify the general firmware flow, button control logic, display behavior, and interaction between the MCU and external modules before testing on the physical prototype.

GitHub was used for project storage and version control. It helps organize firmware files and allows group members to share and update the project source code.

Tool	Purpose
Altium Designer	Schematic design and PCB layout.
STM32CubeIDE	STM32 firmware development, configuration, compilation, and debugging.
ST-Link	Programming and debugging the STM32 through SWD.
Proteus	System-level simulation and firmware behavior verification.
GitHub	Source code management and project file sharing.

Table 2.5. Development Tools and Their Purposes

CHAPTER 3. HARDWARE DESIGN

3.1 Hardware Design Overview

The hardware design of the portable audio player is developed as an integrated embedded system that combines audio playback, user control, power regulation, audio signal routing, output amplification, display, and programming functions on a single circuit board. The complete system schematic is divided into several functional blocks to improve the clarity of the design and to simplify the process of testing and troubleshooting.

The main controller of the system is the STM32F411CEU6 microcontroller. This device is responsible for controlling the operation of the audio player, including reading button inputs, controlling playback functions, selecting the audio signal path, and communicating with the display module. The microcontroller operates from a 3.3 V supply, and external oscillator circuits are included to provide stable clock sources for system operation.

The power supply block provides the required voltage levels for the entire circuit. The system is powered from a battery source, with additional power input support through the USB Type-C connector. The battery voltage is converted to a regulated 5 V supply using a DC-DC converter. This 5 V rail is used by several audio-related components, including the audio module and the amplifier stage. A separate 3.3 V regulator is used to supply the STM32 microcontroller and other low-voltage logic circuits. Decoupling capacitors are placed near the power pins of the main components to reduce supply noise and improve system stability.

The audio source block is based on external audio playback modules that provide left and right audio signals. These signals are then passed to the audio routing stage. An analog multiplexer is used to select the active audio source under the control of the STM32 microcontroller. This allows the system to switch between different audio paths depending on the selected operating mode.

The selected audio signal is sent to the output block, where it can be delivered either to an external audio jack or to the speaker amplifier. The speaker output is driven by an LM386 audio amplifier, which increases the signal power to a suitable level for driving a small speaker. A volume control circuit is included in the signal path to allow adjustment of the output level before amplification.

The user interface block consists of several tactile push buttons and an OLED display. The push buttons are connected to GPIO pins of the STM32 and are used for basic control functions such as play, previous track, next track, mode selection, and volume adjustment. The OLED display communicates with the microcontroller through the I2C interface and is used to show useful information such as the current operating mode, playback status, or other system messages.

For firmware development, the system includes a programming and debugging interface. The SWD header is connected to the programming pins of the STM32, allowing the microcontroller to be programmed and debugged using an external ST-Link programmer. This interface is important during both the development and testing stages of the project.

In general, the hardware design is organized into clear functional blocks: the main controller block, power supply and power management block, audio source block, audio routing and output block, user interface block, and programming/debugging block. This modular design approach makes the schematic easier to understand and provides a practical structure for explaining the operation of the system in the following sections.

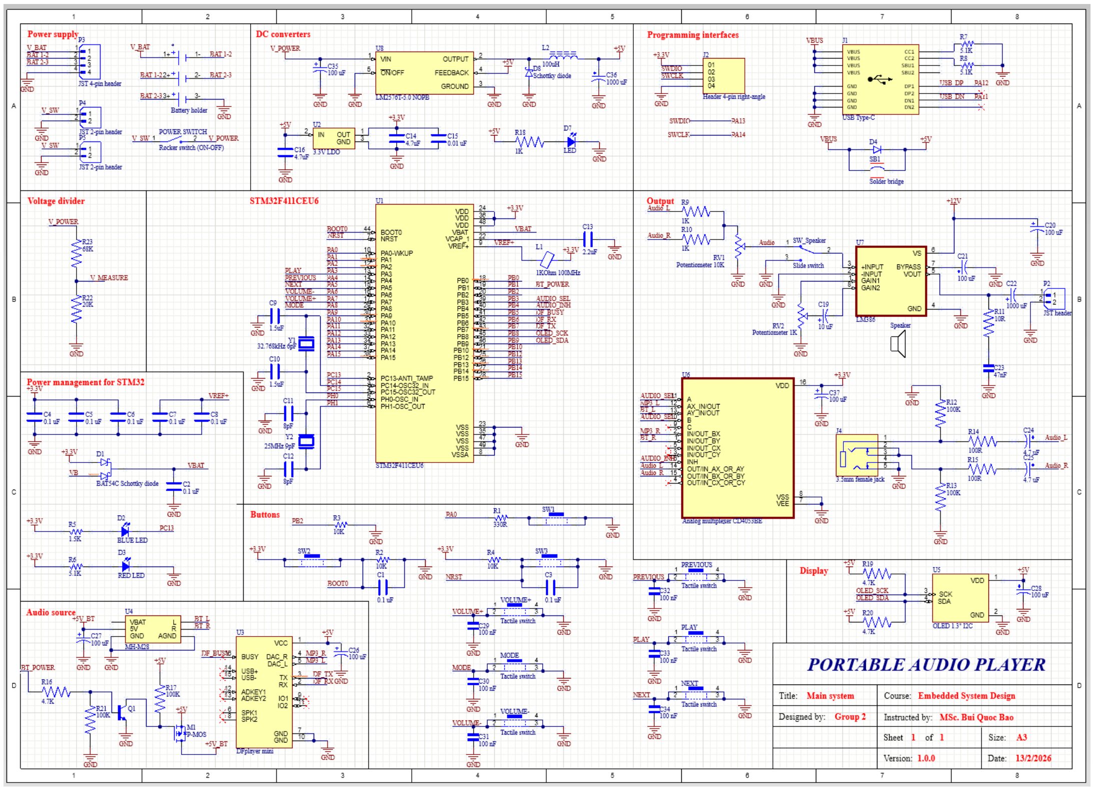

Figure 3.1. Overall Schematic of the MP3 Bluetooth Playback System

3.2 Main Controller Block

The main controller block is built around the STM32F411CEU6 microcontroller, which is used as the central processing unit of the portable audio player. This microcontroller is responsible for controlling the main operation of the system, including reading user inputs, controlling the audio source module, selecting the audio routing path, communicating with the display, and managing the overall playback functions.

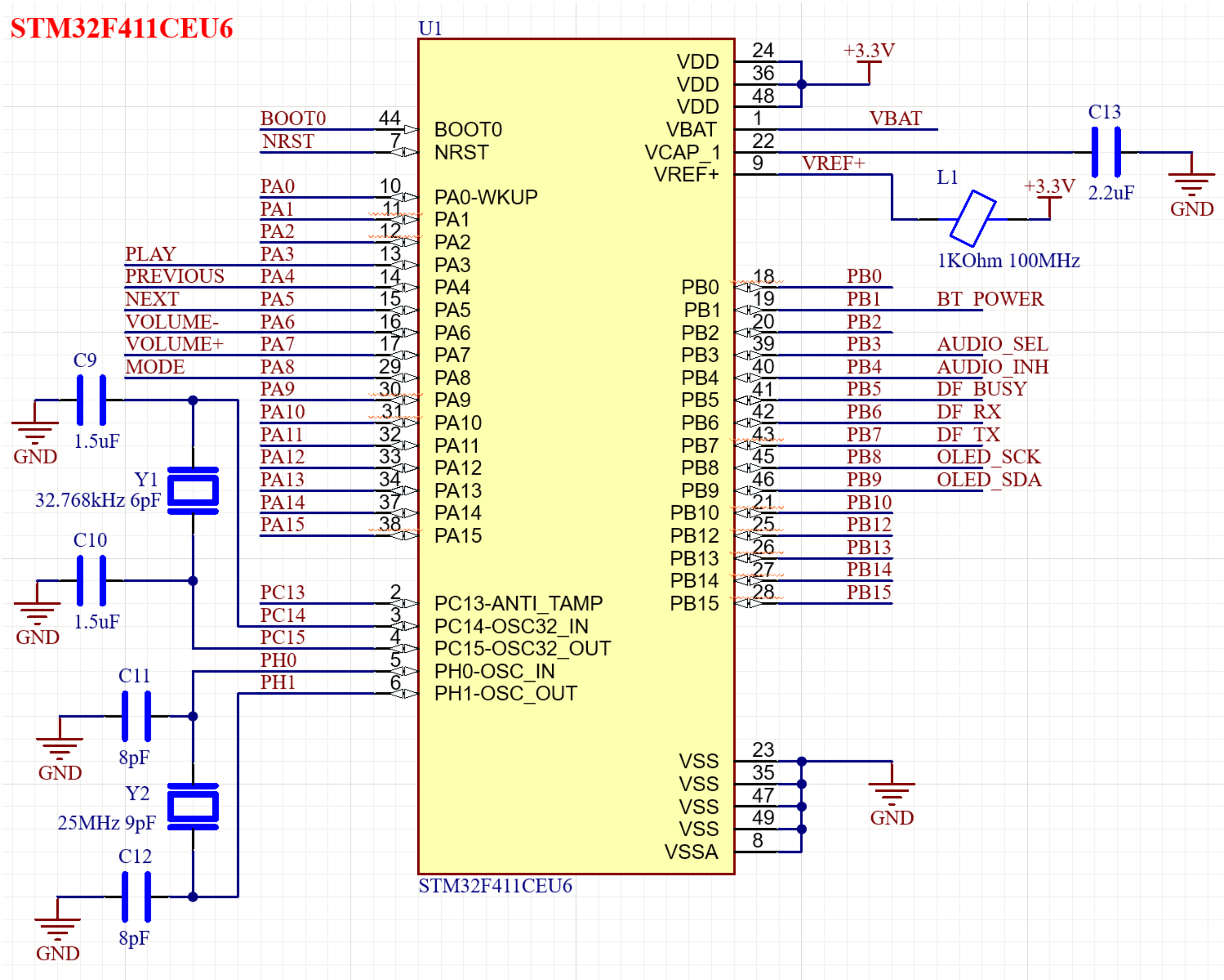

Figure 3.2. STM32F411CEU6 Main Controller Block

The STM32F411CEU6 operates with a 3.3 V supply voltage. In the schematic, the VDD pins of the microcontroller are connected to the regulated 3.3 V power rail, while all VSS pins are connected to the system ground. The analog reference pin VREF+ is also connected to the 3.3 V rail through a filtering component, with a capacitor connected to ground to reduce noise. This helps provide a more stable reference voltage for internal analog functions of the microcontroller. The VCAP pin is connected to an external capacitor, which is required for the internal voltage regulator of the STM32.

The controller uses external clock sources to improve timing accuracy. A 25 MHz crystal oscillator is connected to the PH0 and PH1 oscillator pins to provide the main highspeed external clock. This clock can be used by the STM32 system clock configuration to generate the required operating frequency for the application. A 32.768 kHz crystal oscillator is also connected to the low-speed oscillator pins, allowing the system to support accurate low-frequency timing functions if required. Both oscillator circuits include external load capacitors connected to ground.

The reset and boot configuration pins are also included in this block. The NRST pin is connected to the reset circuit, allowing the microcontroller to be manually or externally reset during development and testing. The BOOT0 pin is used to select the boot mode of the STM32. Under normal operation, BOOT0 is configured so that the microcontroller boots from the internal Flash memory, where the application firmware is stored.

Several GPIO pins of the STM32 are assigned to the user interface buttons. These inputs allow the user to control the audio player functions such as play, previous track, next track, volume adjustment, and mode selection. The button signals are connected to GPIO pins on Port A, as shown in Table 3.1.

Signal	STM32 Pin	Function
PLAY	PA3	Play or pause control
PREVIOUS	PA4	Select previous track
NEXT	PA5	Select next track
VOLUME−	PA6	Decrease volume
VOLUME+	PA7	Increase volume
MODE	PA8	Change operating mode

Table 3.1 Button Input Pin Assignment

The microcontroller also controls the audio source and audio routing circuits through GPIO control signals. The BT_POWER signal is used to control the power state of the Bluetooth audio module. The AUDIO_SEL and AUDIO_INH signals are connected to the analog multiplexer, allowing the firmware to select or disable the audio input path. The DFPlayer module is connected to the STM32 through UART control lines, while the busy signal from the module is connected to a GPIO input so that the controller can detect whether audio is currently being played.

Signal	STM32 Pin	Function
BT_POWER	PB1	Control Bluetooth module power
AUDIO_SEL	PB3	Select audio input path
AUDIO_INH	PB4	Enable or disable audio multiplexer
DF_BUSY	PB5	Detect DFPlayer playback status
DF_RX	PB6	UART transmit signal from STM32 to DFPlayer
DF_TX	PB7	UART receive signal from DFPlayer to STM32

Table 3.2 Audio Control and DFPlayer Interface Pin Assignment

For the display interface, the OLED module communicates with the STM32 using the I2C protocol. The clock and data lines are connected to pins PB8 and PB9 respectively. Pull-up resistors are provided in the display block so that the I2C bus can operate correctly.

Signal	STM32 Pin	Function
OLED_SCK	PB8	I2C clock line
OLED_SDA	PB9	I2C data line

Table 3.3 OLED Display I2C Pin Assignment

Overall, the main controller block provides the digital control required for the complete audio player system. The STM32F411CEU6 receives commands from the user interface, controls the audio modules and routing circuit, communicates with the OLED display, and manages the operating state of the device through firmware. This makes the microcontroller the central coordination unit of the hardware design.

3.3 Power Supply and Management

The power supply and power management block is responsible for receiving power from the battery system and generating the required voltage levels for the whole device. In this design, the circuit provides two main supply rails: 5 V for the audio-related circuits and 3.3 V for the STM32 microcontroller and logic circuits. The block is divided into several smaller parts as described below.

3.3.1 Battery Connection and External Charging Interface

The system is powered by a battery pack connected through JST headers. These headers are used to connect the battery to the external BMS module and charger module. The BMS module provides battery protection, such as over-charge, over-discharge, and over-current protection, while the charger module is used to recharge the battery safely.

The protected battery output is labeled V_BAT. This voltage is used as the main battery supply before it is passed to the rest of the power circuit.

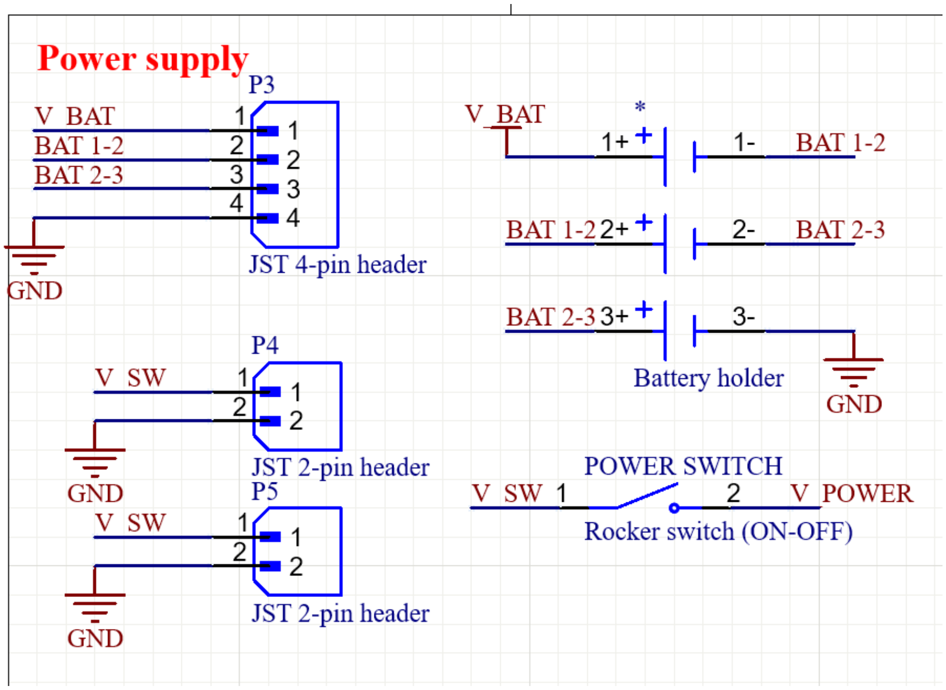

Figure 3.3 Battery Input and Power Switching Circuit

The battery voltage is routed through the main power switch. When the switch is turned on, the battery supply is connected to the main power line labeled V_POWER. This allows the user to turn the whole system on or off manually.

The V_POWER line is then used as the input for the DC converter stage.

3.3.2 The 5V DC Converter and 3.3V LDO regulator

The 5 V rail is generated using the LM2576T-5.0 switching regulator. This regulator converts the battery input voltage into a stable 5 V output.

The converter includes an input capacitor, output inductor, Schottky diode, and output capacitor. The input capacitor helps stabilize the voltage entering the regulator. The inductor and Schottky diode are part of the buck converter operation, while the large output capacitor smooths the output voltage and reduces ripple.

The generated 5 V rail is mainly used to supply higher-power circuits, such as the audio source module and audio amplifier.

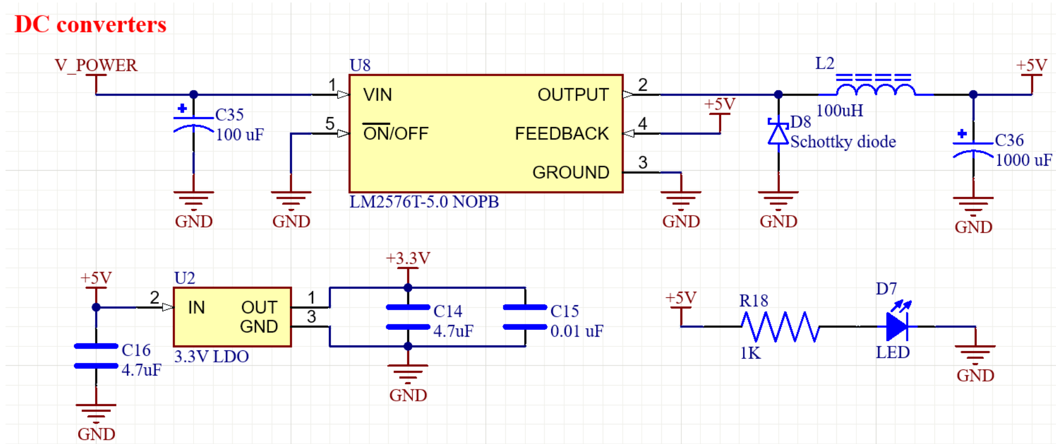

Figure 3.4 5 V DC Converter and 3.3 V LDO Regulator Circuit

The 3.3 V rail is generated from the 5 V supply using a 3.3 V LDO regulator. This rail is required by the STM32F411CEU6 microcontroller and other low-voltage digital circuits.

Input and output capacitors are placed around the LDO to improve voltage stability and reduce noise. Since the STM32 operates at 3.3 V, this regulator is an important part of the system power design.

3.3.3 STM32 Power Management

The STM32 power pins are supplied from the regulated 3.3 V rail. Several decoupling capacitors are connected between 3.3 V and ground near the microcontroller. These capacitors reduce high-frequency noise and provide a stable local power supply during switching operation.

The VREF+ line is also filtered using capacitors to provide a stable reference voltage for the internal analog functions of the STM32. The VBAT pin is supplied through a Schottky diode arrangement, which allows the backup power pin to receive power from the available source while preventing reverse current flow.

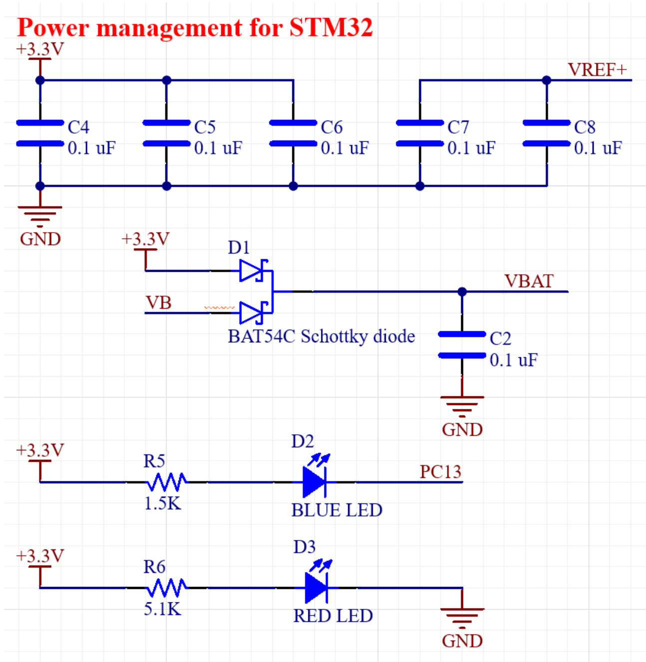

Figure 3.5 STM32 Power Management and Indicator Circuit

Overall, the power supply and power management block provides stable 5 V and 3.3 V supplies for the system, supports battery connection through the BMS and charger module, and includes filtering and indication circuits to improve reliability during operation.

3.4 Audio Source Block

The audio source block provides the audio signals used by the portable audio player. In this design, two audio sources are included: the MH-M28 Bluetooth audio module and the DFPlayer Mini MP3 module. These modules generate analog left and right audio outputs, which are later sent to the audio routing block for source selection.

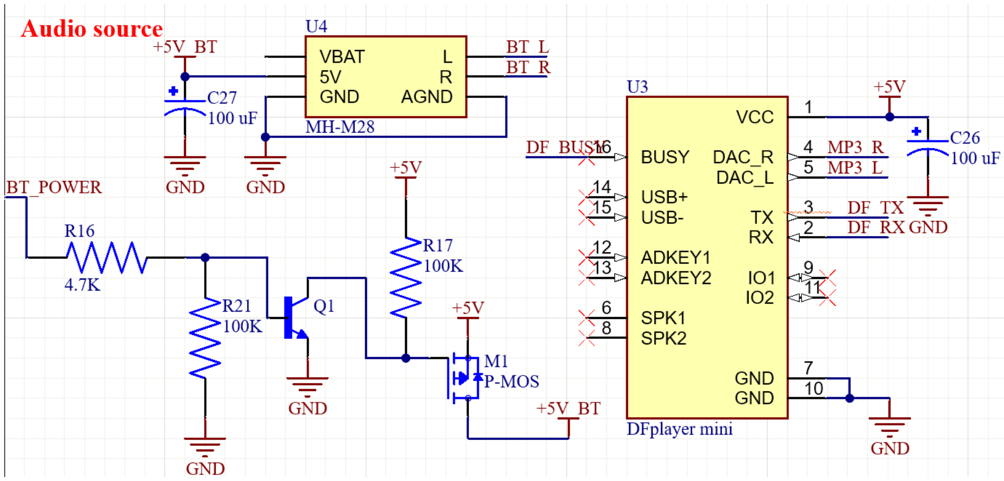

Figure 3.6 **Audio Source Block**

3.4.1 Bluetooth Audio Module

The MH-M28 module is used to provide wireless audio playback through Bluetooth. Its power supply is controlled by the STM32 through the BT_POWER signal. This signal drives a transistor and P-MOSFET switching circuit, allowing the microcontroller to turn the Bluetooth module on or off by controlling the +5V_BT supply line.

When the STM32 enables the BT_POWER signal, the transistor pulls the gate of the P-MOSFET low, turning the MOSFET on and supplying power to the Bluetooth module. When the signal is disabled, the pull-up resistor keeps the P-MOSFET off, disconnecting the module from the 5 V supply. This helps reduce power consumption when Bluetooth audio is not required.

A capacitor is placed near the Bluetooth module supply pins to stabilize the voltage and reduce noise. The module outputs two analog audio signals, labeled BT_L and BT_R, which represent the left and right audio channels.

3.4.2 DFPlayer Mini MP3 Player

The DFPlayer Mini is used as the local MP3 playback source. It is powered from the regulated 5 V rail, with a decoupling capacitor placed close to the module to improve supply stability.

The STM32 communicates with the DFPlayer Mini through a UART interface. The DF_RX and DF_TX lines are used to send and receive control commands, allowing the microcontroller to control functions such as play, pause, next track, previous track, and volume adjustment. The DF_BUSY signal is connected to the STM32 so that the firmware can detect whether the module is currently playing audio.

The DFPlayer provides analog audio outputs from its DAC pins. These outputs are labeled MP3_L and MP3_R, corresponding to the left and right audio channels. These signals are sent to the audio routing section, where they can be selected as the active audio source.

Overall, the audio source block allows the system to support both Bluetooth audio playback and MP3 playback from the DFPlayer Mini. The STM32 controls the operating state of these modules and receives status information from them, while their audio outputs are passed to the next stage for routing and amplification.

3.5 Audio Routing and Output Block

The audio routing and output block is responsible for selecting the active audio source and delivering the selected signal to the final output devices. In this design, the audio signals from the Bluetooth module and the DFPlayer Mini are routed through an analog multiplexer before being sent to the headphone jack and speaker amplifier circuit.

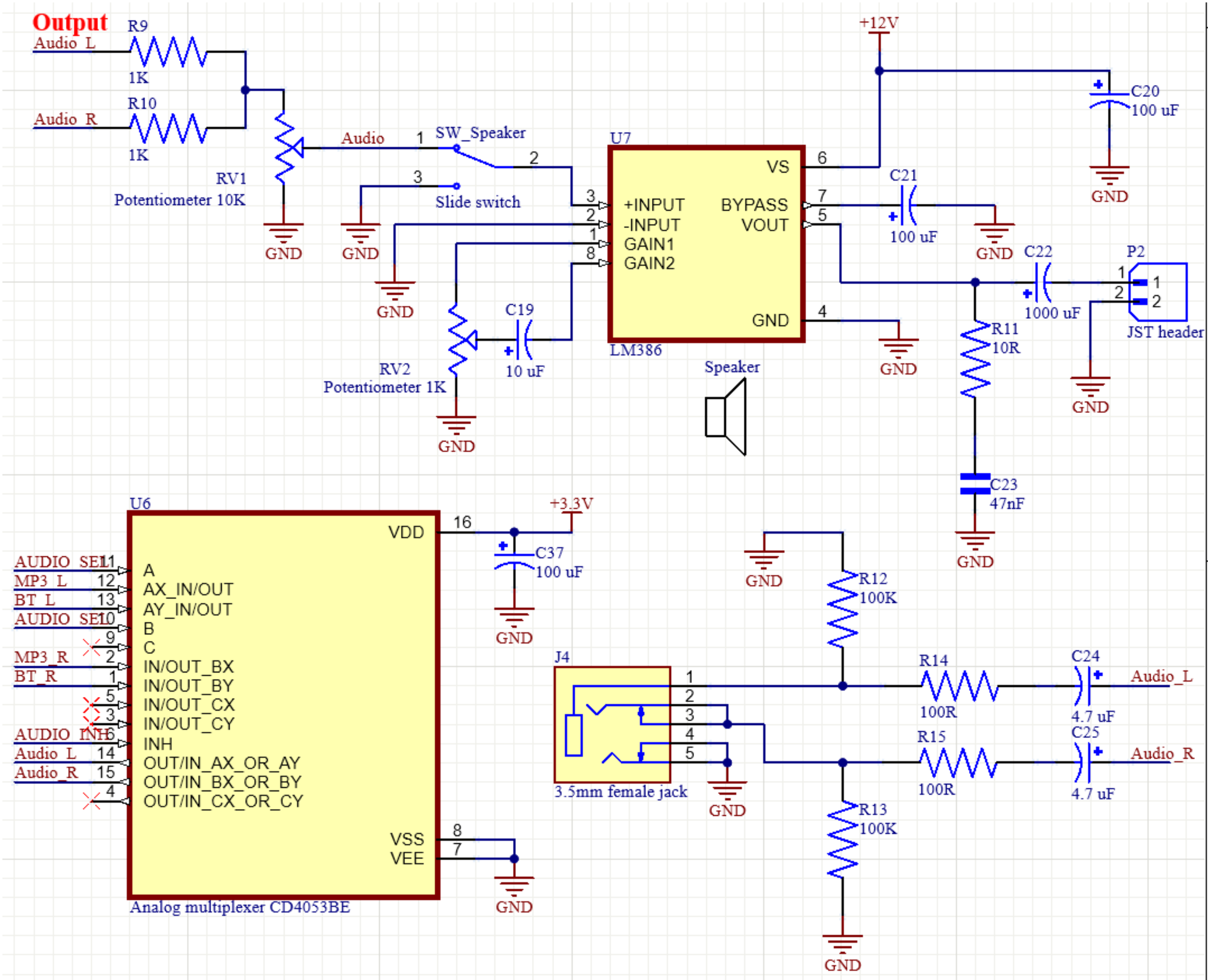

Figure 3.7 Audio Routing and Output Block

3.5.1 Audio Source Selection

The audio source selection circuit uses a CD4053BE analog multiplexer. This IC allows the system to choose between two different stereo audio sources: the Bluetooth audio signals BT_L and BT_R, and the DFPlayer audio signals MP3_L and MP3_R.

The multiplexer is controlled by the STM32 through the signals AUDIO_SEL and AUDIO_INH. The AUDIO_SEL signal selects which audio source is connected to the output path, while AUDIO_INH can be used to enable or disable the audio routing circuit. This allows the firmware to control whether the output audio comes from the Bluetooth module or from the MP3 playback module.

The output of the multiplexer is labeled Audio_L and Audio_R, representing the selected left and right audio channels. These signals are then sent to the external audio jack and the speaker output stage.

3.5.2 External audio jack output

The selected stereo audio signals are connected to a 3.5 mm female audio jack, allowing the user to connect external earphones, headphones, or another audio device. The left and right channels pass through series resistors before reaching the jack, which helps provide basic protection and limits current in the audio path.

Pull-down resistors are also included on the audio lines to define the signal level when no audio source is active and to reduce unwanted noise. Coupling capacitors are placed in series with the audio outputs to block any DC component from reaching the external audio device. As a result, only the AC audio signal is passed to the audio jack.

3.5.3 Speaker Amplifier Configuration

For speaker playback, the left and right audio channels are first combined through resistors to form a single mono audio signal. This is suitable because the speaker output uses one amplifier channel. The combined audio signal then passes through a volume control potentiometer, allowing the user to adjust the signal level before amplification.

A slide switch is included before the amplifier input. This switch allows the speaker path to be enabled or disabled manually. When the speaker path is enabled, the audio signal is passed to the input of the amplifier. When disabled, the amplifier input can be disconnected or pulled to ground, preventing unwanted sound from the speaker.

The speaker output is driven by an LM386 audio amplifier. The LM386 increases the small audio signal from the source module to a level that is suitable for driving a small speaker. The amplifier is supplied from the power rail shown in the schematic, and a large decoupling capacitor is placed near the supply pin to improve stability and reduce power noise.

The amplifier includes external components for gain and output stability. A capacitor and potentiometer are connected between the gain pins of the LM386, allowing the amplifier gain to be adjusted. The bypass capacitor helps reduce noise at the internal bias point of the IC. At the output, a large coupling capacitor is used to block DC voltage while allowing the amplified audio signal to pass to the speaker. A resistor-capacitor network is also connected at the output to improve stability when driving the speaker load.

Overall, the audio routing and output block allows the system to select between Bluetooth audio and MP3 audio, provide stereo output through the 3.5 mm jack, and drive a speaker through the LM386 amplifier. This block therefore forms the final audio path of the portable audio player.

3.6 User Interface

The user interface block allows the user to control the portable audio player and receive basic system information. This block consists of tactile push buttons for input control and an OLED display for visual feedback.

3.6.1 Push Buttons

The system uses several tactile switches connected to the GPIO pins of the STM32 microcontroller. These buttons are used for the main playback and menu control functions, including play/pause, previous track, next track, volume up, volume down, and mode selection.

Each button is connected so that pressing the switch pulls the corresponding input signal to ground. In normal operation, the input pin is kept at a logic high level by a pull-up connection, either through hardware or the internal pull-up configuration of the STM32. Therefore, the firmware can detect a button press by checking for a low logic level on the corresponding GPIO pin.

Small capacitors are connected across several button inputs to reduce contact bouncing. Mechanical switches do not change state cleanly when pressed or released, so these capacitors help smooth the signal before it is read by the microcontroller. Additional software debouncing can also be implemented in the firmware to improve reliability.

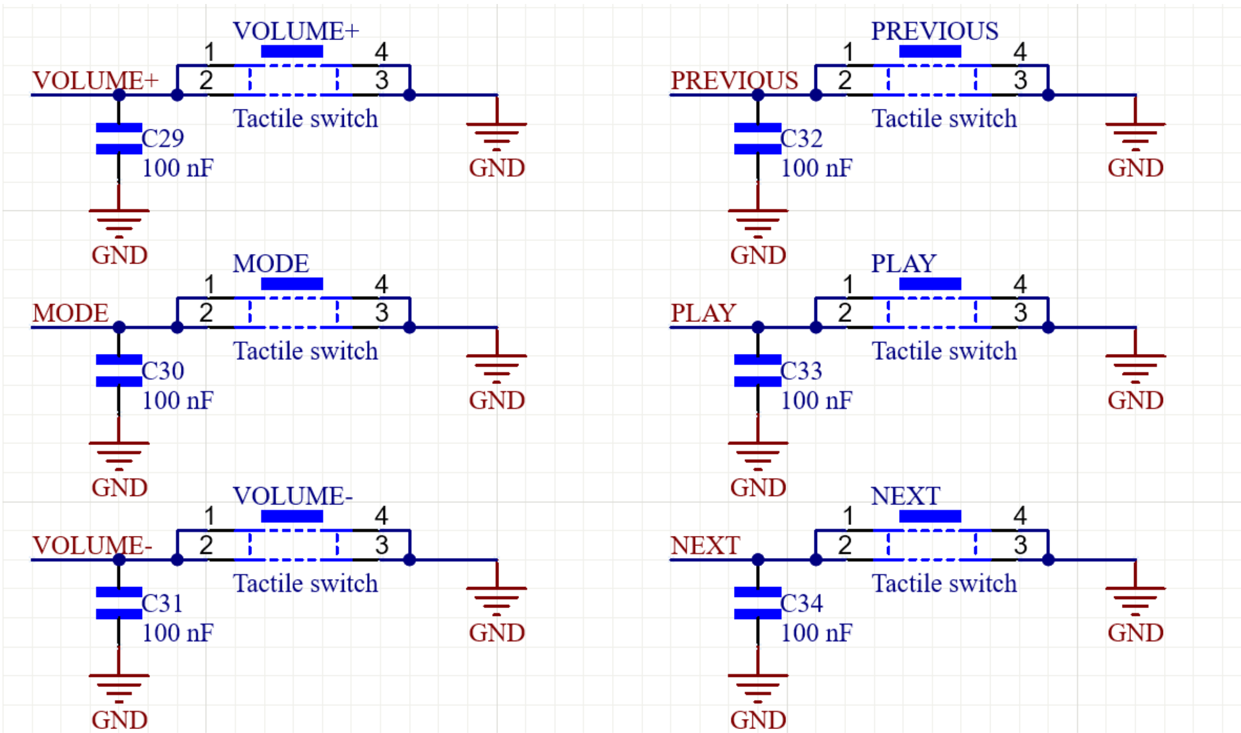

Figure 3.8 Push Button Input Circuit

Besides the playback control buttons, the design also includes BOOT0 and NRST buttons. The BOOT0 button is used to select the STM32 boot mode during startup, while the NRST button allows the microcontroller to be manually reset during development and testing.

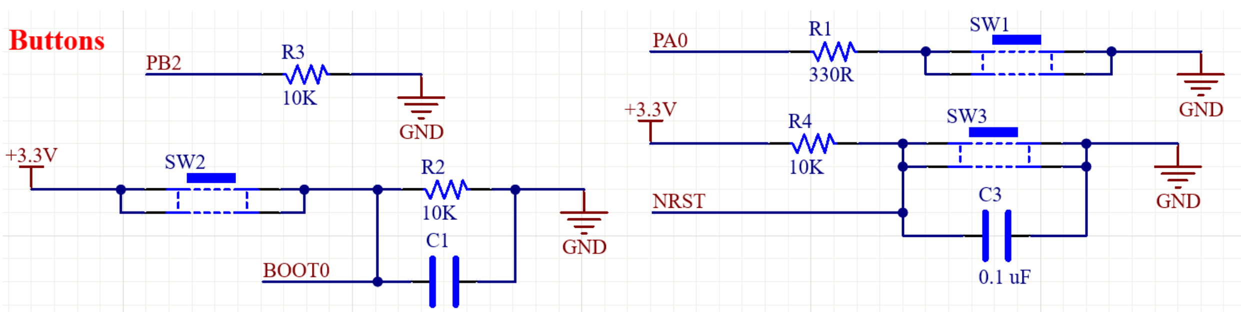

Figure 3.9 Boot Mode and Reset Circuit

3.6.2 OLED Display

The display section uses a 1.3-inch I2C OLED module. The OLED is used to show important information to the user, such as the current playback mode, track status, volume level, or other system messages.

The OLED communicates with the STM32 through the I2C interface, using two signal lines: OLED_SCK for the clock line and OLED_SDA for the data line. Pull-up resistors are connected to these lines so that the I2C bus can operate correctly. A decoupling capacitor is also placed near the OLED power supply pin to stabilize the supply voltage and reduce noise.

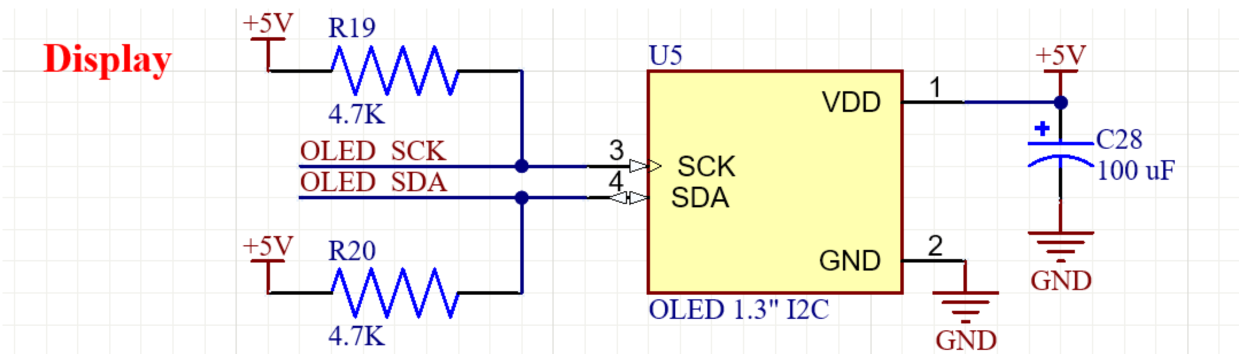

Figure 3.10 OLED Display Interface Circuit

Overall, the user interface block provides both input and output interaction for the system. The buttons allow the user to control the audio player, while the OLED display gives visual feedback about the operating state of the device.

3.7 Programming and Debugging Interface

The programming and debugging interface is included to support firmware development, testing, and troubleshooting of the STM32 microcontroller. This block mainly consists of the SWD programming header and the USB Type-C connector.

The SWD header provides the required connections for programming and debugging the STM32 using an external ST-Link programmer. The header includes 3.3 V, GND, SWDIO, and SWCLK. The SWDIO and SWCLK signals are connected to pins

PA13 and PA14 of the STM32, respectively. Through this interface, the firmware can be uploaded to the microcontroller, and debugging functions such as breakpoints, variable monitoring, and program tracing can be performed during development.

The USB Type-C connector is used as an external USB interface for the system. The VBUS and GND pins provide the USB power connection, while the USB_DP and USB_DN lines are connected to the STM32 USB data pins. The CC1 and CC2 pins are connected through pull-down resistors, allowing the connector to be detected correctly as a USB device connection.

A diode and solder bridge are also included between VBUS and the system 5 V rail. This provides flexibility in the power configuration, allowing the USB input to be used as a possible 5 V source when required, while the diode helps prevent reverse current flow.

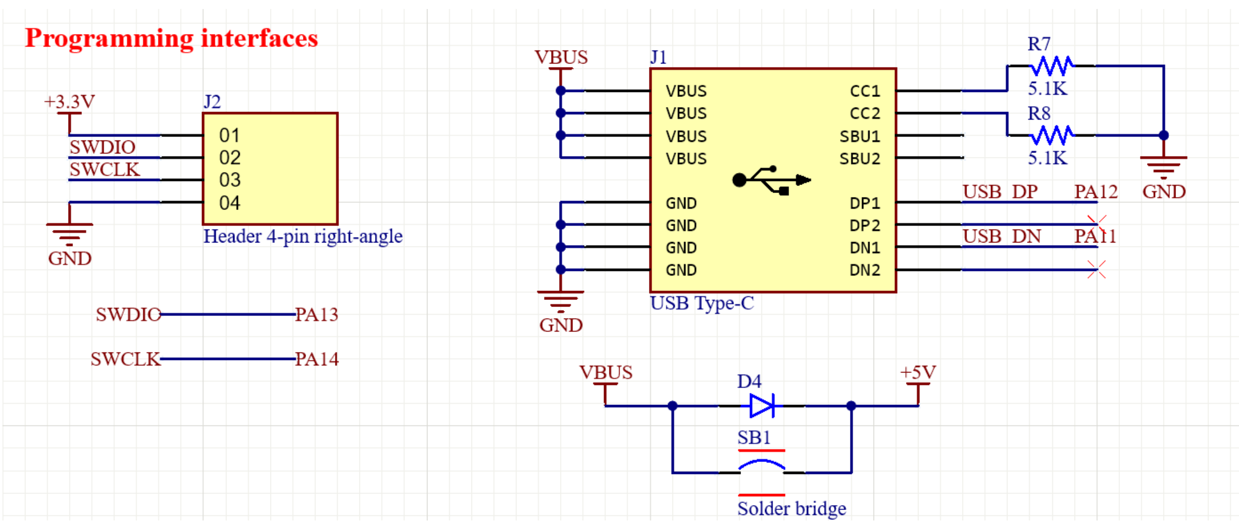

Figure 3.11 Programming and USB Type-C Interface Circuit

Overall, this block provides the necessary interfaces for programming, debugging, and possible USB connection of the device during development and operation.

CHAPTER 4. FIRMWARE ARCHITECTURE

4.1 Firmware Design Objective

The firmware of the MP3 / Bluetooth Playback System is designed using a modular and layered architecture. The main objective is to separate the application control logic from the hardware implementation. This makes the firmware easier to understand, maintain, and modify when the hardware connection or peripheral driver is changed.

The system is implemented using a super-loop architecture, meaning the firmware repeatedly executes service functions inside the main loop. Time-dependent tasks, such as battery monitoring, OLED updating, and DFPlayer busy-pin monitoring, are handled using HAL_GetTick() - based timing.

The firmware supports the following main functions:

- MP3 playback control using DFPlayer Mini.
- Bluetooth audio mode switching.
- Button input handling with debounce.
- OLED user interface display.
- Battery voltage and percentage monitoring.

4.2 Layered Firmware Architecture

The firmware is divided into four main layers:

- Application Layer
- Board Support Package Layer
- Low-Level Driver / Asset Layer
- STM32 HAL and Hardware Layer

The firmware is organized using a layered architecture to separate the system-level control logic from hardware-specific implementation. In this design, the application layer does not directly control GPIO pins, UART commands, ADC readings, or OLED drawing functions. Instead, it calls high-level BSP APIs such as audio control, button event reading, battery monitoring, and display updating.

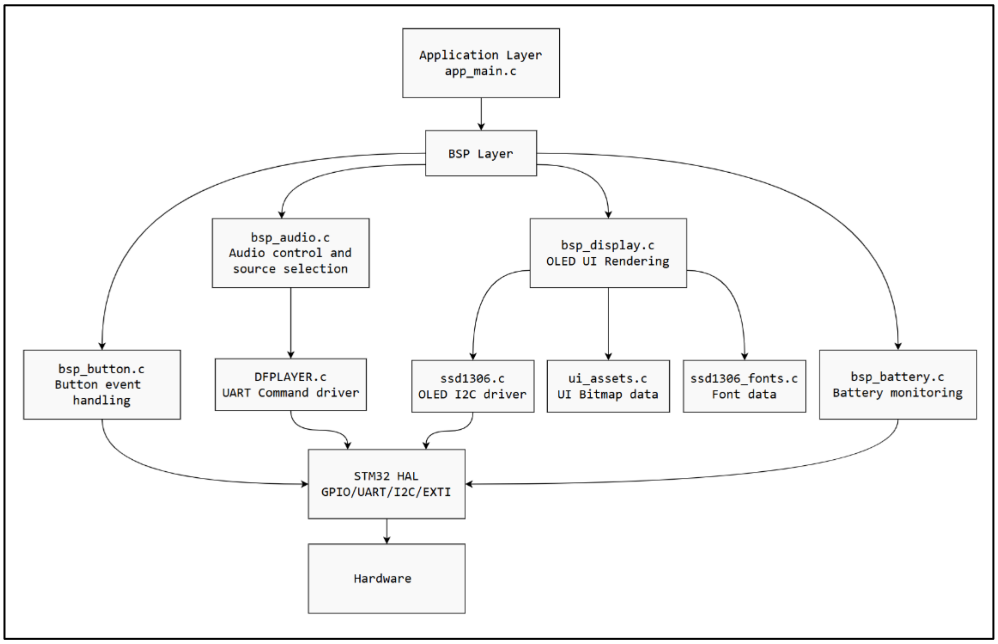

Figure 4.1 Firmware Layered Architecture

Table 4.1 summarizes the responsibility of each firmware layer. The main purpose of this separation is to keep the application layer simple and focused on system behavior, while hardware access and device-specific operations are handled by lower layers.

Table 4.1 Layer Description

Layer	Main files	Responsibility
Application Layer	app_main.c	Coordinates the whole system, handles button events, and calls periodic service functions.

BSP Layer	bsp_button.c, bsp_audio.c, bsp_display.c, bsp_battery.c	Provides high-level board functions and hides GPIO, ADC, UART, and display details from the application.
Low-Level Driver Layer	DFPLAYER.c, ssd1306.c	Provides device-specific communication functions.
Data / Asset Layer	ui_assets.c, ssd1306_fonts.c	Stores bitmap images and font data for OLED rendering.
HAL / Hardware Layer	main.c, STM32 HAL	Initializes peripherals and provides low-level hardware access.

Overall, this firmware architecture allows the project to be explained at system-design level instead of function-by-function implementation level. The application layer describes what the system should do, while the BSP and driver layers define how each hardware module performs its operation.

4.3 Main Firmware Execution Flow

The firmware starts from the STM32 generated main.c file. First, the HAL library, system clock, GPIO, I2C, UART, and ADC peripherals are initialized. After that, the user application is initialized by calling App_Init().

After initialization, the firmware enters an infinite loop and repeatedly calls App_Loop().

App_main implementation

```
void App_Init(UART_HandleTypeDef *huart, ADC_HandleTypeDef *hadc) {
    BSP_Button_Init();
    BSP_Audio_Init(huart);
```

```
    HAL_Delay(1000u);


    BSP_Battery_Init(hadc);

    BSP_Display_Init();

    BSP_Display_FirstDraw();

}


void App_Loop(void) {

    BSP_ButtonEvent_t evt;


    BSP_Audio_Service();

    BSP_Battery_Service();


    while (BSP_Button_PopEvent(&evt) != 0u) {

        App_HandleEvent(evt);

    }


    BSP_Display_Service();

}
```

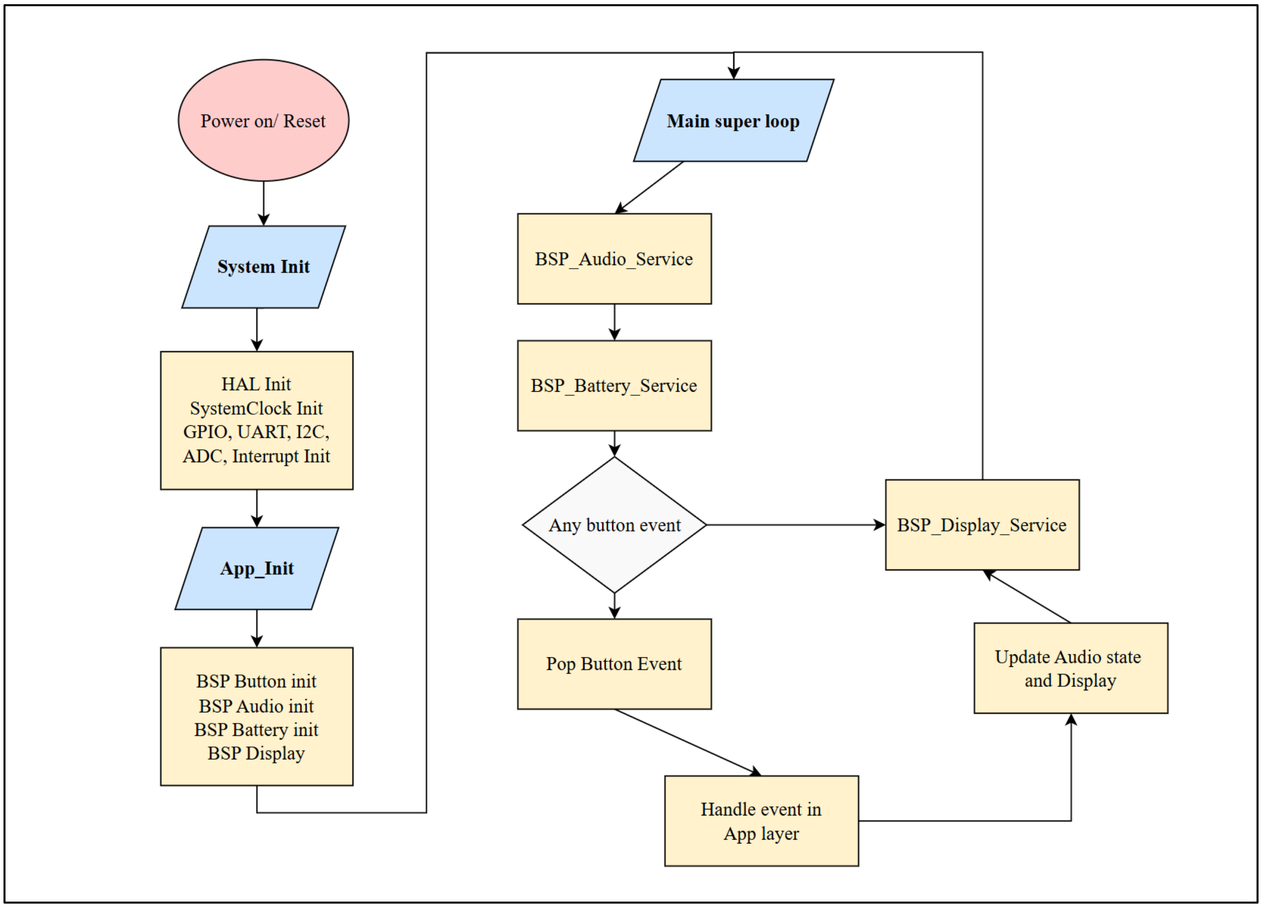

Figure 4.2. Main Firmware Flowchart

The main loop performs three important operations:

1. **Audio service:** Checks the DFPlayer busy pin and performs automatic next-track handling.
2. **Battery service**: Periodically samples the battery voltage through ADC and updates the estimated battery percentage.
3. **Button event processing**: Reads button events from the queue and converts them into audio control actions such as play/pause, next, previous, volume up/down, and mode switching.
4. **Display service**: Updates the OLED only when necessary and animates the playback wave during MP3 playback.

4.4 Event Handling Mechanism

The button system is designed as an interrupt-driven input system with an event queue. When a button is pressed, the GPIO EXTI interrupt is triggered. The HAL callback function HAL_GPIO_EXTI_Callback() maps the pressed GPIO pin to a high-level button event. A debounce time is applied to prevent multiple false triggers caused by mechanical bouncing.

After debounce, the event is pushed into a small ring buffer. The application layer does not process the button directly inside the interrupt. Instead, it later reads the event from the queue inside the main loop.

App_HandleEvent implementation

```
static void App_HandleEvent(BSP_ButtonEvent_t evt) {

    switch (evt)

    {

        case BSP_BUTTON_EVT_PLAY_TOGGLE:

            BSP_Audio_PlayToggle();

            BSP_Display_ResetWaveAnim();

            BSP_Display_RefreshAll();

            break;


        case BSP_BUTTON_EVT_PREV_TRACK:

            BSP_Audio_Prev();

            BSP_Display_ResetWaveAnim();

            BSP_Display_RefreshAll();

            break;


        case BSP_BUTTON_EVT_NEXT_TRACK:
```

```
            BSP_Audio_Next();
            BSP_Display_ResetWaveAnim();
            BSP_Display_RefreshAll();
            break;

        case BSP_BUTTON_EVT_MODE_TOGGLE:
            BSP_Audio_ModeToggle();
            BSP_Display_ResetWaveAnim();
            BSP_Display_RefreshAll();
            break;

        case BSP_BUTTON_EVT_VOL_DOWN:
            BSP_Audio_VolDown();
            BSP_Display_RefreshAll();
            break;

        case BSP_BUTTON_EVT_VOL_UP:
            BSP_Audio_VolUp();
            BSP_Display_RefreshAll();
            break;

        case BSP_BUTTON_EVT_NONE:
        default:
            break;
    }
}
```

The supported button events are:

Button Event	System Action
Play/Pause	Toggle DFPlayer playback
Previous	Go to previous track
Next	Go to next track
Mode	Switch between MP3 SD mode and Bluetooth mode
Volume Up	Decrease DFPlayer volume
Volume Down	Increase DFPlayer volume

Table 4.2: Supported events

4.5 BSP Module Responsibilities

4.5.1 Button BSP

The Button BSP is responsible for converting physical button presses into high-level firmware events. It handles GPIO EXTI interrupts, debounce timing, and event buffering.

The application layer only receives clean events such as PLAY_TOGGLE, NEXT_TRACK, or MODE_TOGGLE. Therefore, the application does not need to know the actual GPIO pin of each button.

Main responsibilities:

- Configure logical mapping between buttons and events.
- Handle EXTI callback.
- Apply software debounce.
- Store events in a ring buffer.
- Allow the application to pop events safely in the main loop.

4.5.2 Audio BSP

In MP3 mode, the Audio BSP sends UART commands to the DFPlayer Mini through the low-level DFPLAYER.c driver. It supports play/pause, next track, previous track, and volume adjustment.

In Bluetooth mode, the Audio BSP switches the analog audio path to the Bluetooth module and disables DFPlayer control actions. It also controls the Bluetooth enable pin.

Main responsibilities:

- Initialize DFPlayer Mini through UART.
- Set initial volume.
- Select MP3 or Bluetooth audio source.
- Control analog switch selection.
- Mute/unmute the audio path during switching.
- Track current playback state.
- Monitor DFPlayer busy pin.
- Automatically move to the next track when playback finishes.

4.5.3 Display BSP

The Display BSP controls the OLED user interface. It uses the SSD1306 driver and bitmap assets to render different display screens.

The display shows different UI states depending on the selected audio source and playback state. In MP3 mode, it displays playback information such as track number, volume, battery percentage, and play/pause animation. In Bluetooth mode, it displays the Bluetooth screen and battery status.

To reduce I2C traffic, the display module caches the previous system state and updates only the changed areas when possible.

Main responsibilities:

- Initialize the SSD1306 OLED.
- Draw MP3 pause screen.
- Draw MP3 play screen and wave animation.
- Draw Bluetooth screen.
- Display track number, volume, and battery percentage.
- Refresh only changed UI regions.

4.5.4 Battery BSP

The Battery BSP monitors the battery pack voltage using the ADC. The battery voltage is measured through a resistor divider, converted into pack voltage, and mapped to a percentage value for display.

The module does not update continuously every loop. Instead, it updates periodically every fixed interval to reduce unnecessary ADC usage and stabilize the displayed value.

Main responsibilities:

- Read ADC values from the battery voltage divider.
- Average multiple ADC samples.
- Convert ADC value to battery pack voltage.
- Estimate battery percentage.
- Apply a simple low-pass filter to smooth the displayed result.

CHAPTER 5. HARDWARE IMPLEMENTATION

After completing the schematic design, the hardware of the portable audio player was implemented as a custom PCB. The purpose of this stage was to convert the designed circuit into a physical prototype that can be assembled, powered, programmed, and tested. The implemented hardware includes the battery holder, power supply circuit, STM32 main controller, audio source modules, audio routing circuit, speaker amplifier, OLED display, push buttons, audio jack, and programming interface.

The PCB was designed with consideration for both electrical operation and mechanical arrangement. Large components such as the battery holder, speaker, OLED display, buttons, potentiometers, and audio jack were positioned according to the expected user interaction and available board area. Smaller components such as resistors, capacitors, diodes, regulators, and ICs were placed near their related functional blocks to simplify routing and improve system reliability.

5.1 PCB Layout

The PCB layout was designed based on the schematic presented in the previous chapter. The board has a rectangular shape with a size of approximately 150 mm × 140 mm. A 2-layer PCB structure was used, where the top and bottom layers were used for signal routing, power distribution, and ground copper pours.

The placement of components was arranged according to the function of each block. The battery holder was placed in the top left corner area of the PCB because it occupies a large mechanical space. The speaker was placed in the top right corner area so that the sound output can be directed outward from the board. The OLED display and push buttons were placed on the user-accessible side of the PCB to make the device easier to operate.

The STM32 microcontroller was placed near the center of the control section to reduce the routing distance to the buttons, OLED display, audio control signals, and programming header. The power supply components were grouped close to the power input and switch section to keep the high-current paths short. The audio routing and output components were placed near the audio jack, potentiometers, and speaker amplifier to reduce the length of audio signal traces.

Ground copper pours were applied on both layers to provide a stable ground reference for the system. This also helps reduce noise and improves the return path for power and signal currents.

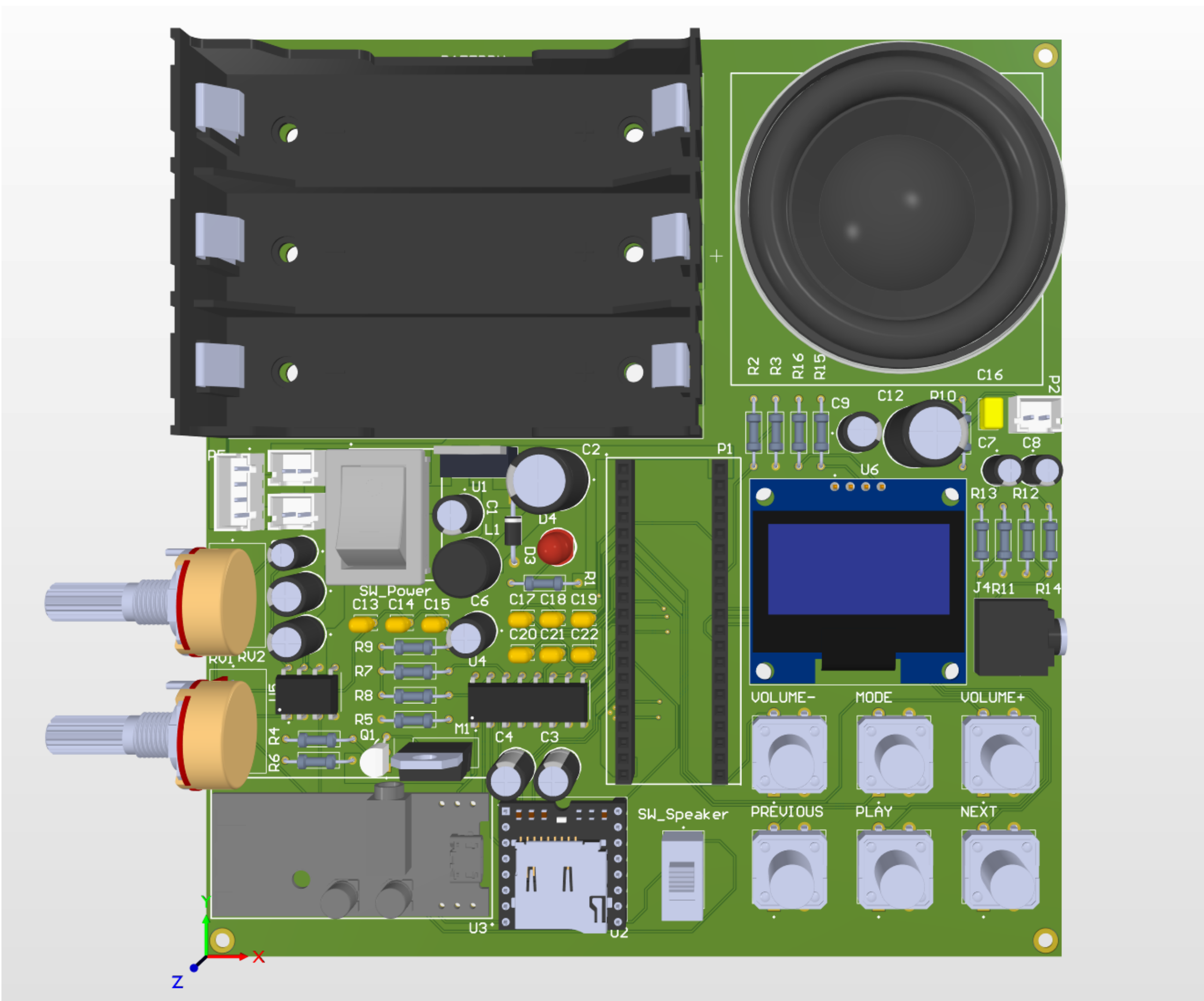

Figure 5.1 3D PCB Layout of the MP3 Bluetooth Playback System

5.2 PCB Design Rules and Manufacturing Parameters

The PCB design rules were selected based on the capability of the PCB manufacturer and the electrical requirements of the system. Since the circuit includes both low-current digital signals and higher-current power/audio paths, different trace widths were used depending on the function of each net.

Small signal traces such as GPIO, UART, I2C, button signals, and control lines were routed using a trace width of 0.25 mm. Copper pours were used for power lines such as V_BAT, V_POWER, 5 V, and amplifier supply lines. This helps reduce voltage drop and improves current-carrying capability.

The PCB uses vias with a diameter of 0.6 mm and a drill size of 0.3 mm. Through-hole component pads were selected according to the mechanical size of each component pin. The clearance between traces, pads, and copper pours was set to 2.54 mm to satisfy the manufacturing requirement and reduce the risk of short circuits.

The main PCB manufacturing parameters are summarized in the following table

Parameter	Value
PCB size	**150 mm × 150 mm**
PCB shape	**rectangular**
Number of layers	**2 layers**
Board thickness	**1.6 mm**
Copper thickness	**1 oz**
Minimum signal trace width	**0.2 mm**
Minimum clearance	**2.54 mm**
Via pad diameter	**0.6 mm**
Via drill diameter	**0.3 mm**

Solder mask color	**Black**
Mounting hole size	**M2**

Table 5.1 PCB Design Rules and Manufacturing Parameters

5.3 PCB Fabrication and Assembly

After the PCB layout was completed and checked using the Design Rule Check function, the manufacturing files were generated for PCB fabrication. These files included the Gerber files for the copper layers, solder mask, silkscreen, board outline, and the NC drill file for the holes and vias.

Before fabrication, the design was reviewed to ensure that the board outline, mounting holes, component footprints, trace clearances, and silkscreen labels were correct. Special attention was given to the footprints of large and mechanical components such as the battery holder, speaker, OLED display, audio jack, potentiometers, push buttons, JST connectors, and module headers.

After the PCB was fabricated, all components were soldered manually by hand. The assembly process was carried out step by step, starting with the power supply section. This allowed the main voltage rails to be tested before connecting the STM32 microcontroller and other modules. After the power section was completed, the STM32 controller circuit, crystal oscillators, reset and boot components, SWD programming header, audio circuits, display, buttons, connectors, and audio modules were assembled.

During hand soldering, components with polarity such as electrolytic capacitors, diodes, LEDs, and the battery connector were checked carefully before applying power. After assembly, the PCB was visually inspected to identify possible solder bridges, incorrect component placement, or weak solder joints. This step helped reduce hardware faults before the initial power-up and system testing.

5.4 Hardware Implementation Result

The final hardware prototype successfully integrates the main functional blocks of the portable audio player into a single PCB. The implemented board includes the battery holder, power supply circuit, STM32 controller, audio source modules, analog audio routing circuit, speaker amplifier, OLED display, push buttons, audio jack, and programming interface.

The hardware implementation provides a complete platform for firmware development and system-level testing. After the board was assembled and verified, the firmware could be developed to control the audio sources, read user inputs, update the OLED display, and manage the operating modes of the portable audio player.

CHAPTER 6. CONCLUSION AND FUTURE WORK

6.1 Conclusion

This project presented the design and implementation of an MP3 Bluetooth Playback System using the STM32F411CEU6 microcontroller. The system was designed to support two playback modes: MP3 playback from a microSD card and Bluetooth audio playback from a paired device. It also includes OLED display feedback, button-based control, audio source switching, speaker output, 3.5 mm audio output, battery monitoring, and portable power operation.

The project followed a complete embedded system design process. First, the system requirements and main functions were defined. Then, suitable components were selected, including the STM32F411CEU6, DFPlayer Mini, Bluetooth audio receiver module, OLED display, CD4053BE analog multiplexer, LM386 amplifier, and voltage regulation circuits. After that, the schematic was designed by dividing the system into functional blocks. The PCB layout was then created to integrate the complete design into a compact board.

The firmware was organized using a modular structure. The main program initializes the peripherals and runs the main loop, while separate modules handle input processing, playback control, OLED rendering, battery monitoring, and application-level event handling. This approach makes the firmware easier to understand, debug, and extend.

The system was verified using Proteus simulation and breadboard testing. The prototype demonstrated the main functions of the system, including MP3/Bluetooth playback control, OLED display update, button interaction, audio source switching, and audio output. The PCB has also been fabricated and is being assembled for final testing. Overall, the project achieved its main objective of designing and implementing a functional embedded audio playback system from schematic and firmware design to prototype validation and PCB implementation.

6.2 Limitations

Although the main system functions were achieved, several limitations remain.

First, audio noise is still present in the current prototype. This issue may be caused by power supply noise, breadboard wiring, grounding problems, or interference between digital control signals and analog audio paths. Since audio circuits are sensitive to noise, this problem requires further improvement in both hardware design and PCB layout.

Second, the PCB still requires complete post-assembly validation. After soldering, each block must be tested carefully, including the power supply, STM32 programming interface, OLED display, buttons, DFPlayer Mini, Bluetooth module, audio multiplexer, amplifier circuit, speaker output, and audio jack.

Third, the audio quality was evaluated mainly through practical listening tests. More detailed audio measurements, such as output noise level, distortion, frequency response, and maximum output power, were not performed due to limited equipment.

Fourth, the current design focuses mainly on system functionality and embedded system integration. Mechanical enclosure design, advanced battery charging, battery protection, and product-level audio optimization are not included in the current implementation.

6.3 Future Improvements

Future improvements can focus on making the system more reliable, compact, and suitable for practical use.

The first improvement is audio noise reduction. This can be achieved by improving PCB grounding, separating analog and digital signal paths, adding proper decoupling capacitors near each module, filtering the power rails, and keeping sensitive audio traces away from switching regulators and digital signals.

The second improvement is completing full PCB-level testing. The assembled PCB should be tested step by step, starting from power rail verification, then MCU programming, display operation, button input, MP3 playback, Bluetooth playback, audio source switching, and amplifier output.

The third improvement is better battery management. A future version of the system can include a charging circuit, battery protection circuit, more accurate battery percentage estimation, and low-battery warning on the OLED display.

The fourth improvement is enhancing the user interface. The OLED screen can be used to show more detailed information such as current mode, track number, playback state, volume level, battery percentage, and Bluetooth connection status.

The final improvement is mechanical integration. A compact enclosure can be designed to protect the PCB and make the system easier to use as a portable device.

Overall, the current design provides a functional foundation for an embedded MP3 Bluetooth playback system. With further improvements in audio noise reduction, PCB testing, battery management, and enclosure design, the system can be developed into a more complete and reliable portable audio player.

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